Examinations Sample Questions With Solutions

1. (a)

   (i) Express $\frac{2-\sqrt{2}}{3+\sqrt{2}}$ in the form $a + b\sqrt{2}$ where a and b are rational numbers.

   (ii) Find x and y if $\frac{3-2i}{1+4i} = \frac{x+iy}{1+i}$.

   (iii) Express $0.3\overline{25}$ in the form $\frac{a}{b}$ where a and b are rational numbers.

   (b) Let X and Y be subsets of a universal set U such that $X \subset Y$,

   (i) express $(X \cup (X \cup Y)')'$ in its simplest form.

   (ii) simplify $((X' \cap Y)' \cap Y)'$ as far as possible.

   (c) The polynomial $p(x) = ax^3 + 5x^2 - bx - 6$ leaves a remainder of -12 when divided by $x - 1$. If further, $x + 2$ is a factor of the polynomial $p(x)$,

   (i) find the values of a and b.

   (ii) hence factorize the polynomial $p(x)$ completely.

   (iii) solve also the inequality $p(x) \geq 0$.

2. (a) Given that $\mathbf{R}$, the set of real numbers is the universal set, $A = [-12, 13)$ and $B = [-4, \infty)$, find the following sets:

   (i) $A' \cap (B - A)$.

   (ii) $(A - B) \cup (B - A)$.

   (iii) $(A \cup B)'$

   (b)

   (i) Let $f(x) = \frac{x}{2x-1}$ and $g(x) = \frac{3}{x+2}$ be two functions. Find $(fog)(x)$.

   (ii) Express $\frac{3+2i}{4-i}$ in the form $a + ib$ where a and b are rational numbers.

   (iii) Solve the equation $(\log_3 x)^2 - \log_3 x = 2$ for real values of x.

   (c)

   (i) Given that α and β are the roots of the equation $3x^2 + x - 5 = 0$, find an equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

   (ii) By completing the square, express the function $f(x) = -3x^2 - x + 5$ in the form $-3(x + p)^2 + q$ where p and q are real numbers.

   (iii) Hence sketch the graph of the function $f(x) = -3x^2 - x + 5$.

3. (a)

   (i) Solve the inequality $\frac{2x}{x+3} \leq \frac{1}{2}$.

(ii) Express $\frac{2}{5+2\sqrt{3}}$ in the form $a+b\sqrt{3}$ where a and b are real numbers.

(iii) Solve the equation $2^x - 9 + 8(2^{-x}) = 0$.

(b)

(i) Given the rational function $f(x) = \frac{2-3x}{2x+1}$;
Find $f^{-1}(x)$, the inverse of $f(x)$.

(ii) Sketch the graph of the function $f(x) = 2x^3 - 3x^2 - 3x + 2$.

(c)

(i) Given that α and β are the roots of the equation $2x^2 - 3x - 7 = 0$, find an equation whose roots are α^2 and β^2.

(ii) Sketch the graph of the quadratic function $f(x) = 2x^2 - 3x - 5$.

(iii) Let $f(x) = \frac{2x-1}{x+1}$ and $g(x) = \frac{x+1}{2}$ be functions. Find $(fog)^{-1}(x)$

4. (a)

(i) Let $\mathbb{R}$, the set of real numbers be the universal set. If $A = [-7, 3)$ and $B = \{x : x \leq 10, x \in \mathbb{R}\}$, Find the set $B - A$ and display your answer on a number line.

(ii) Express $0.\overline{38}$ in the form $\frac{a}{b}$ where a and b are integers and $b \neq 0$.

(b)

(i) Simplify $\sqrt{32} + \sqrt{18}$ giving your answer in the form $a\sqrt{2}$ where a is an integer.

(ii) Simplify $\frac{\sqrt{32}+\sqrt{18}}{3+\sqrt{2}}$ giving your answer in the form $x + y\sqrt{2}$, where x and y are rational numbers.

(c)

(i) Express $[(A \cap B)' \cup (A - B)]'$ in its simplest form where A and B are sets.

(ii) Express $\frac{(-2+3i)^2}{1+2i}$ in the form $a + ib$ where $a, b \in \mathbb{R}$.